(Please write your Enrolment No. immediately) Enrolment No. _____

SUPPLEMENTARY MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, May 2024

Paper Code: DA-304T **Subject: Statistics, Statistical Modelling and Data Analytics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	COs
1 a	What is descriptive analysis?	2	CO1
b	What is the difference between simple and multiple linear regression?	2	CO2
c	Explain binomial distribution.	2	CO1
d	Explain z-test.	2	CO2
e	What is the different between null and alternate hypothesis testing?	2	CO2
2a	Elaborate the Gauss-Markov theorem in detail.	5	CO1, CO2
2b	Consider the following dataset that shows the relationship between the number of hours spent on studying and the exam score for a class-	5	CO2

Hours	10	20	30	40	50	60	70
Score	6	5	7	5	15	10	12

i. Calculate the means of hours and exam score.
ii. Determine the slope and the intercept for the simple linear regression equation y= mx+b.
iii. Write the equation for the regression line.
iv. Predict the score for the student who studied for 15 hours.

3a	Find the mean, median, mode, range and standard deviation of the following dataset-	5	CO1

Marks	10	15	20	25	30	35	40
No. of Students	45	17	18	34	7	18	23

3b	Elaborate on the need of data visualization. Why do we need it in statistical analysis? Discuss the various ways through which it can be done?	5	CO1
4a	State and discuss the binomial probability distribution theorem. 20 coins are tossed simultaneously. Find the probability of getting at least 12 heads.	5	CO1, CO2
4b	A man is known to speak truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.	5	CO2

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6th Semester, April, 2024

Paper Code: DA-304T **Subject: Statistics, Statistical Modelling and Data Analytics**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	COs
1 a	Explain null and alternate hypothesis.	2	CO1
b	Briefly explain logistic regression.	2	CO2
c	What is the probability that a leap year selected at random will contain 53 Sundays?	2	CO1
d	Briefly explain ANOVA testing.	2	CO2
e	What is the different between one-tailed and two-tailed hypothesis testing?	2	CO2
2a	Find the mean, median, mode, range and standard deviation of the following dataset-	5	CO1

Marks	0-10	11-20	21-30	31-40	41-50
No. of Students	2	7	11	6	1

2b	Explain data visualization. What is the need of it in statistical analysis? elaborate on the advantages and disadvantages of it. Discuss the various techniques that are used for the task of data visualization.	5	CO1
3a	Differentiate between discreet random variable and continuous random variable. If the mean and the variance of a binomial distribution are 4 and 2 respectively, then find the probability of exactly 2 successes.	5	CO2
3b	A firm produces electric bulbs that have burning life normally distributed with mean 800 hours and standard deviation of 40 hours. Find that probability that a random sample of 16 bulbs will have average burning life of less than 775 hours. {P(0<z<2.5) =0.4938}	5	CO2
4a	Explain the Gauss-Markov theorem in detail.	5	CO1, CO2
4b	Consider the following dataset that shows the relationship between the number of years of experience and the salary of the employee	5	CO2

Experience (in years)	2	4	6	8	10
Salary (in lakhs)	5	8	10	15	20

i. Calculate the means of years of experience and mean of salaries.
ii. Determine the slope and the intercept for the simple linear regression equation y= mx+b.
iii. Write the equation for the regression line.
iv. Predict the salary for an employee with 15 years of experience.